

Virtual Try-On & Measurement Architecture Flow

This architecture enables merchants to upload clothing products, generate AI-enhanced product visuals, allow customers to virtually try on garments, capture body measurements, and complete purchases through messaging platforms such as WhatsApp and Telegram.

Merchant Onboarding and Subscription Verification

The process begins when a merchant accesses the platform hosted on Vercel.

1. The merchant logs in through the **API Gateway & Auth** service.
2. The API checks the merchant's **Account & Subscription Status** stored in the Merchant Data database.
3. The **Subscription Gatekeeper** continuously validates whether the merchant's Hubtel subscription is active.
4. If payment is overdue:
 - Product inventory becomes hidden.
 - Virtual try-on functionality is disabled.
 - API access is restricted.

This ensures that only paying merchants can use platform features.

Product Upload and AI Image Enhancement

When a merchant adds a new clothing item:

1. The merchant uploads garment photos through the **Product Catalog**.
2. The request is sent to the **Image Metadata Service**.
3. The image is forwarded by the **Photo Upload Handler** to **Google Gemini 2.5 Flash**.
4. Gemini receives instructions from the **Prompt Templates Engine**, which provides prompts for:
 - Background removal.
 - Image cleaning.
 - Garment isolation.
 - Virtual try-on preparation.
5. Gemini generates an optimized product image.
6. The processed image metadata is returned to the **Image Metadata Service**.
7. Product information and image references are stored in **Inventory Metadata**.

At this stage, the merchant has AI-enhanced product images ready for customer interaction.

Customer Virtual Try-On Experience

When a customer wants to see how a garment might look on a person:

1. The customer opens the **Virtual Try-On Interface**.

2. The customer selects a preferred model style through the **Try-On Model Selector**.
3. The selector retrieves the appropriate prompt template from the **Prompt Engine**.
4. The garment image is sent to Gemini together with:
 - Selected model style.
 - Garment image.
 - Try-on prompt template.
5. Gemini generates a virtual try-on image showing the selected garment on the chosen model.
6. The generated image is displayed back in the Virtual Try-On Interface.

This provides a realistic visualization without requiring expensive 3D rendering infrastructure.

Measurement Capture Process

To improve fit recommendations:

1. The customer opens the **Measurement Scanner UI**.
2. The browser activates the device camera.
3. **MediaPipe** runs directly inside the browser.
4. MediaPipe detects and tracks 33 body landmarks in real time.
5. Landmark coordinates are converted into measurement data.
6. The resulting measurements are sent to the **Measurement Storage Service**.
7. Measurements are stored in the **Measurements** database.

Because MediaPipe runs on the user's device:

- No mobile app is required.
- No dedicated backend compute is required.
- Processing costs remain extremely low.
- User privacy is improved because raw video stays on the device.

Basket Creation and Checkout

After selecting products:

1. The customer adds items to the basket.
2. The **Basket & Checkout** component creates an order request.
3. The request is sent to the **Basket Link Generator**.
4. The generator:
 - Packages selected products.
 - Attaches customer measurements.
 - Creates a shareable checkout link.

The basket now contains both the selected items and the customer's body measurement profile.

Social Commerce Distribution

Instead of forcing customers through a traditional e-commerce checkout:

1. The Basket Link Generator can export the order to:
 - **WhatsApp.**
 - **Telegram.**
2. The customer or merchant shares the generated basket link.
3. The merchant receives the order through their preferred communication channel.
4. Order references are stored in **Order History**.

This approach aligns with social-commerce purchasing behavior common in many emerging markets.

Payment and Subscription Management

Merchant subscription payments are handled by Hubtel.

1. The merchant pays the monthly fee of GHS 75/month.
2. Hubtel processes:
 - Mobile Money payments.
 - Debit/Credit card payments.
3. Hubtel updates:
 - Subscription status records.
 - Payment state information.
4. The Subscription Gatekeeper immediately reflects the new status.
5. Platform access is automatically granted or revoked based on payment status.

This creates a fully automated Software-as-a-Service billing model.

End-to-End Example

A typical journey looks like this:

1. Merchant uploads a dress.
2. Gemini cleans and optimizes the product image.
3. Customer visits the merchant's store.
4. Customer chooses a model style.
5. Gemini generates a virtual try-on image.
6. Customer scans their body using MediaPipe.
7. Measurements are stored.
8. Customer adds the dress to their basket.
9. Basket Generator attaches measurement data.
10. A WhatsApp checkout link is created.
11. Merchant receives the order.
12. Order details are saved in Order History.
13. Merchant continues using the platform as long as the Hubtel subscription remains active.

Architectural Advantages

Area	Benefit
AI Processing	Gemini Flash provides low-cost image generation and enhancement.
Measurement Capture	MediaPipe runs locally with no server cost.
Hosting	Vercel provides scalable frontend hosting.
Backend	Supabase simplifies authentication, storage, and APIs.
Billing	Hubtel supports Ghana-specific payment methods.
Commerce	WhatsApp and Telegram enable social selling.
Scalability	Most compute occurs client-side or through managed services.
Cost Efficiency	Minimal infrastructure and operational expenses.

The overall design is a lightweight, subscription-based social-commerce platform that combines AI-generated virtual try-ons, browser-based body measurement capture, and messaging-driven checkout while keeping infrastructure costs very low.